SIMATS ENGINEERING

ASSESSMENT TOOL 1:

Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**

Question Count: 4

Code of Conduct

I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded

Set 2 - Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

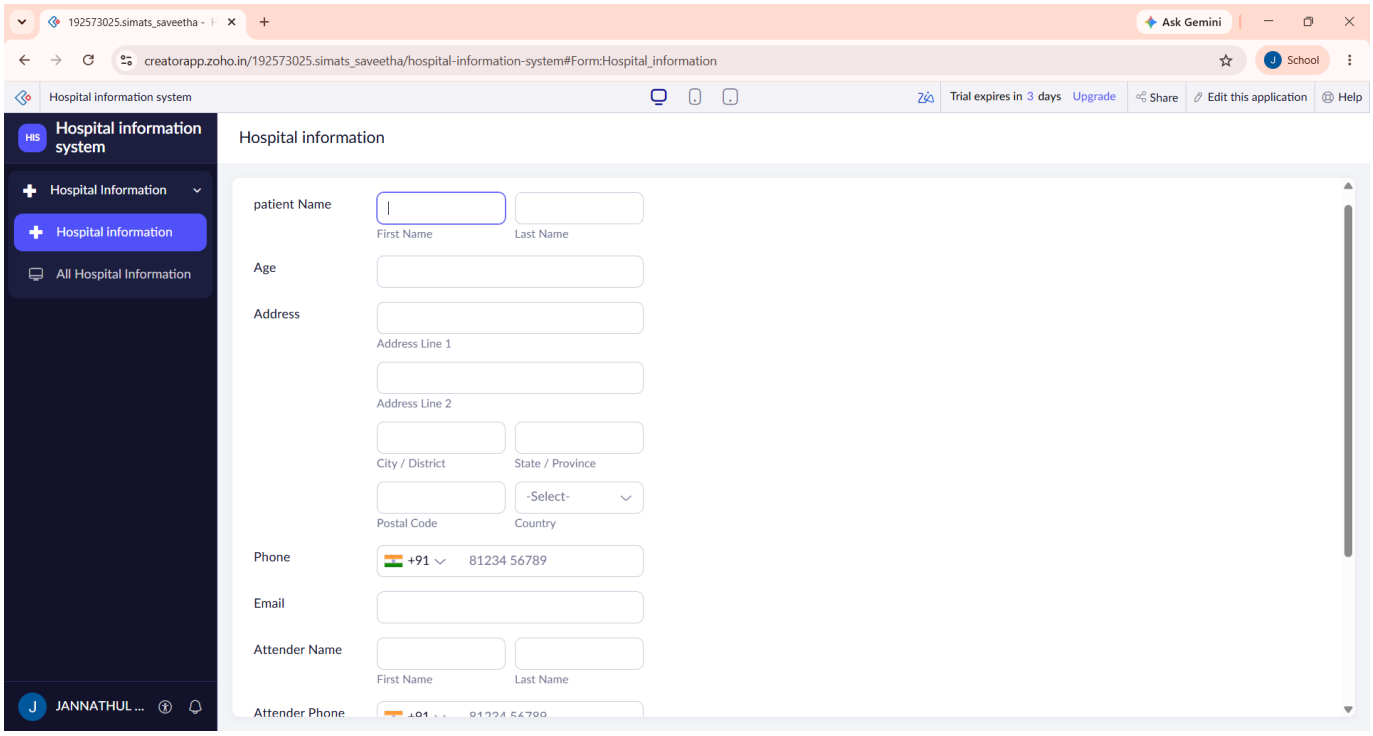
Experiment 1: Hospital Information System using SaaS

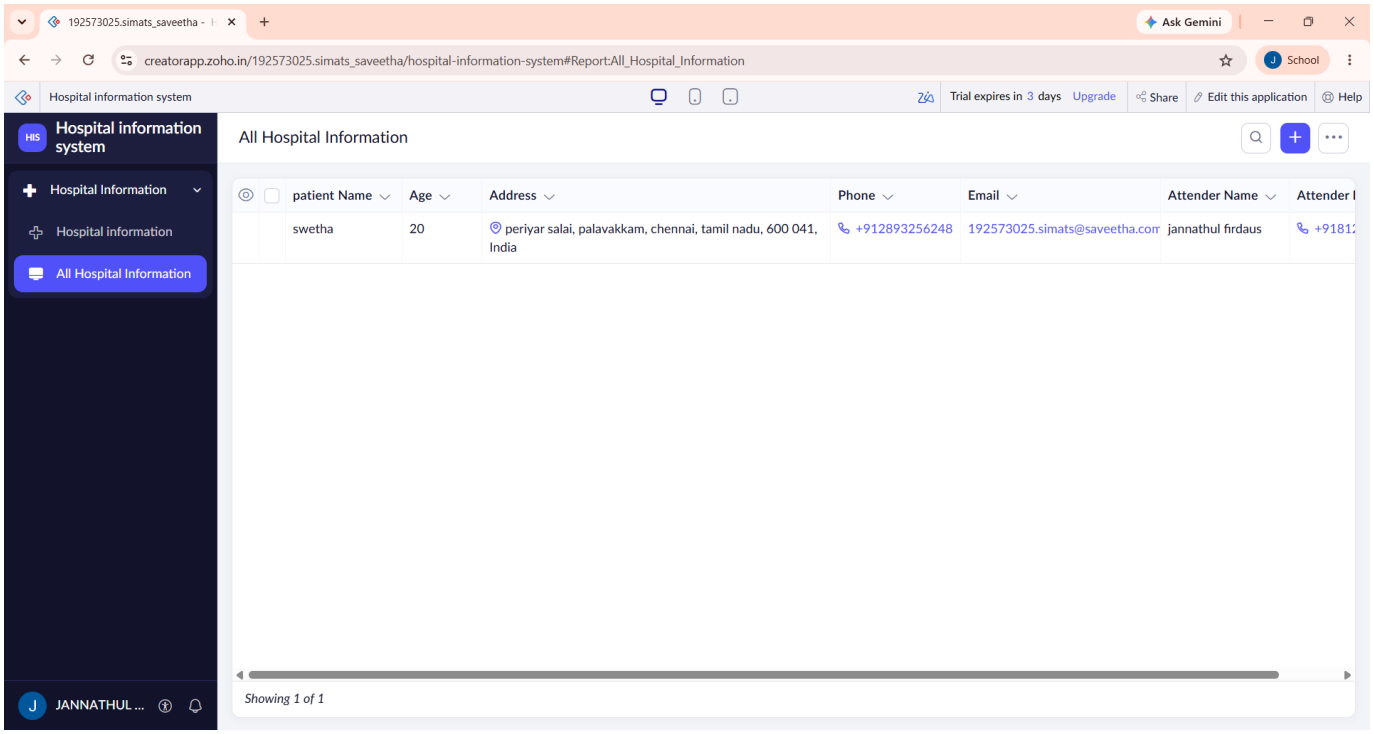
Aim:

To create a cloud-based Hospital Information System using a cloud service provider to manage patient, attender, doctor, and disease details.

Algorithm:

1. Create a cloud database using Google Sheets.
2. Open Google AppSheet and create an application using the database.
3. Add fields such as patient name, age, address, phone, email, attender name, attender phone, doctor name, and disease.
4. Create a form to enter patient details.
5. Save and view the patient records.
6. Test the application.





Result:

Thus, a cloud-based Hospital Information System was created to store and manage patient information using SaaS.

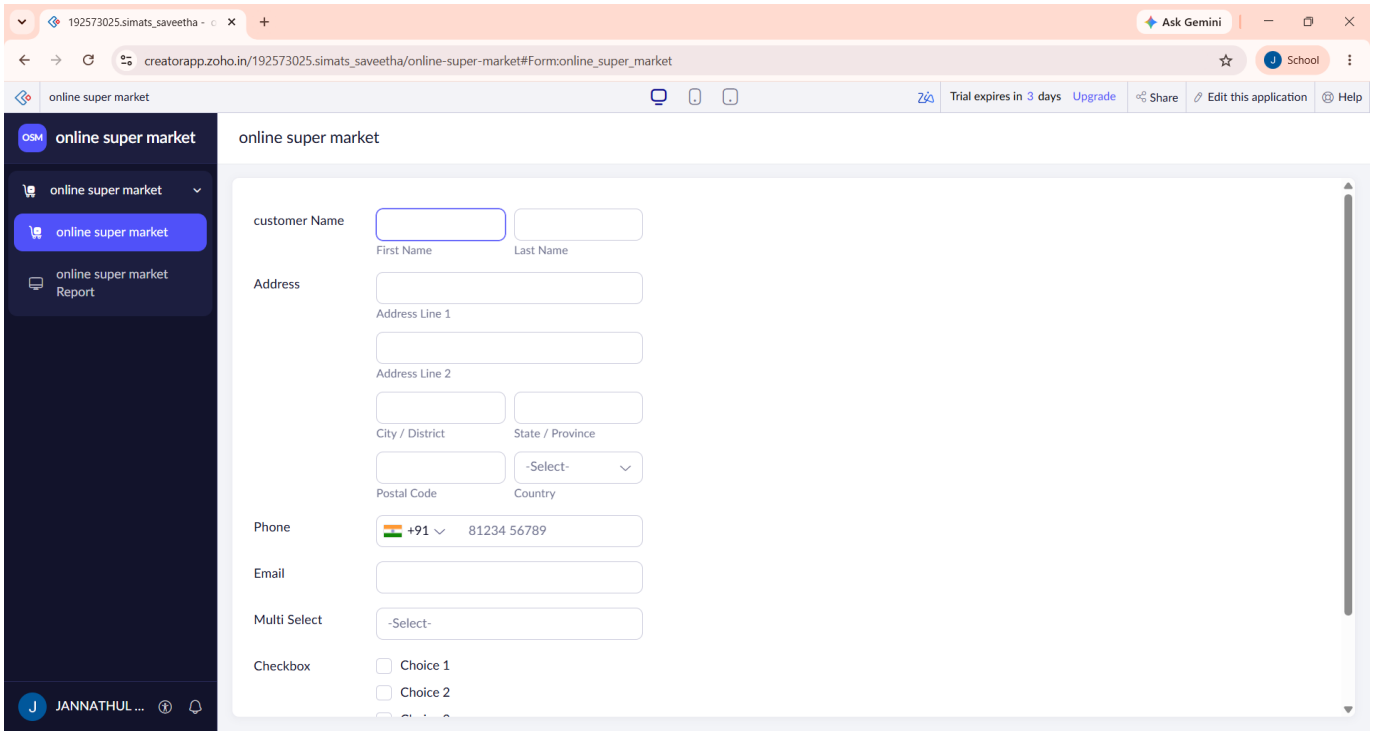
Experiment 2: Online Supermarket Application using SaaS

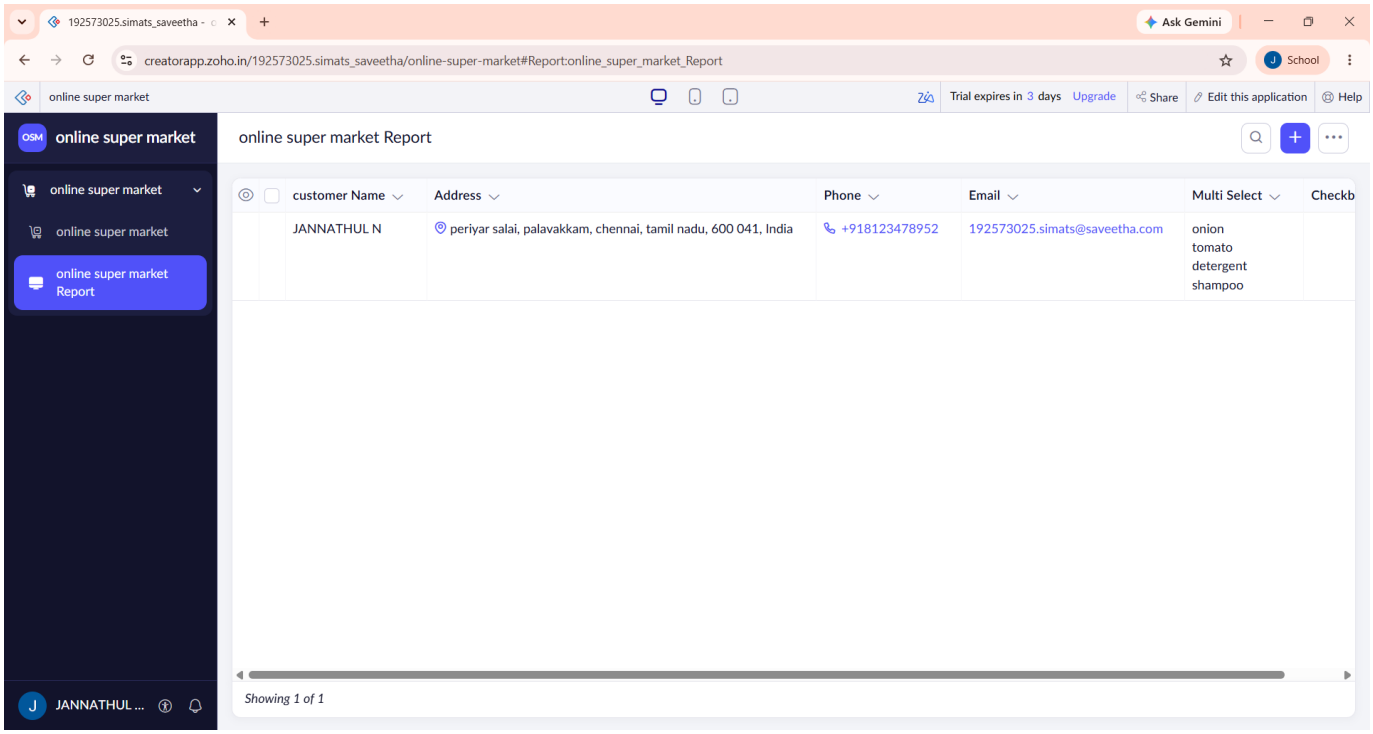
Aim:

To create a cloud-based Online Supermarket Application to manage customer details, items, quantities, and item classifications.

Algorithm:

1. Create a cloud database using Google Sheets.
2. Create an application using Google AppSheet.
3. Add customer fields such as name, address, phone, and email.
4. Create an item classification form with item name, category, price, and quantity.
5. Create an order form to select items and enter quantities.
6. Save and view customer and order details.
7. Test the application.





Result:

Thus, a cloud-based Online Supermarket Application was created to manage customer details, item classifications, and orders using SaaS.

Experiment 3: Create a Virtual Machine using a Type-2 Hypervisor

Aim:

To create a virtual machine using VMware Workstation Pro with 3 CPUs, 10 GB RAM, and 10 GB storage.

Algorithm:

1. Open VMware Workstation Pro.

2. Select Create a New Virtual Machine.

3. Choose the Ubuntu operating system.

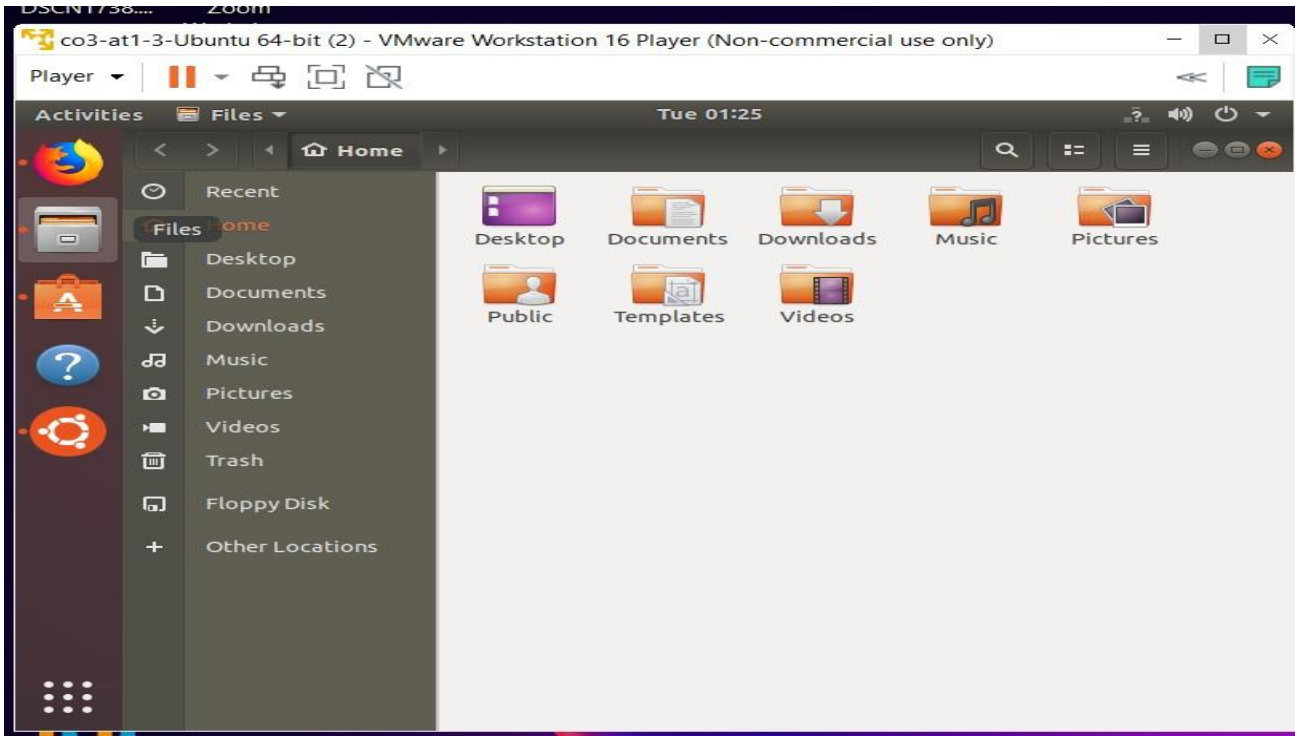
4. Set the virtual machine name and location.

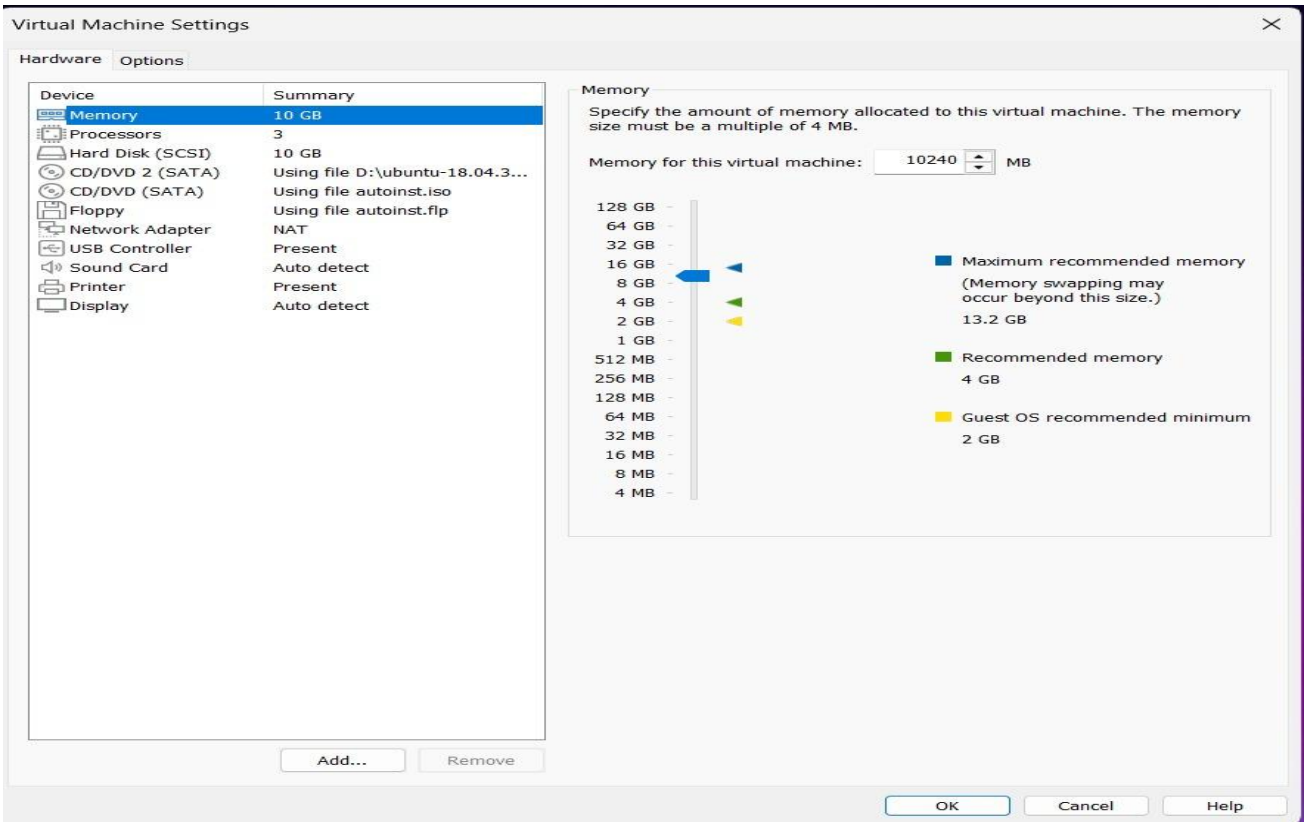
5. Configure the processor with 3 CPU cores.

6. Allocate 10 GB RAM and a 10 GB virtual disk.

7. Create the virtual machine and install Ubuntu.

8. Start the VM and verify its configuration.





Result:

Thus, a virtual machine was created using VMware Workstation Pro with virtual disk storage, and Ubuntu was installed.

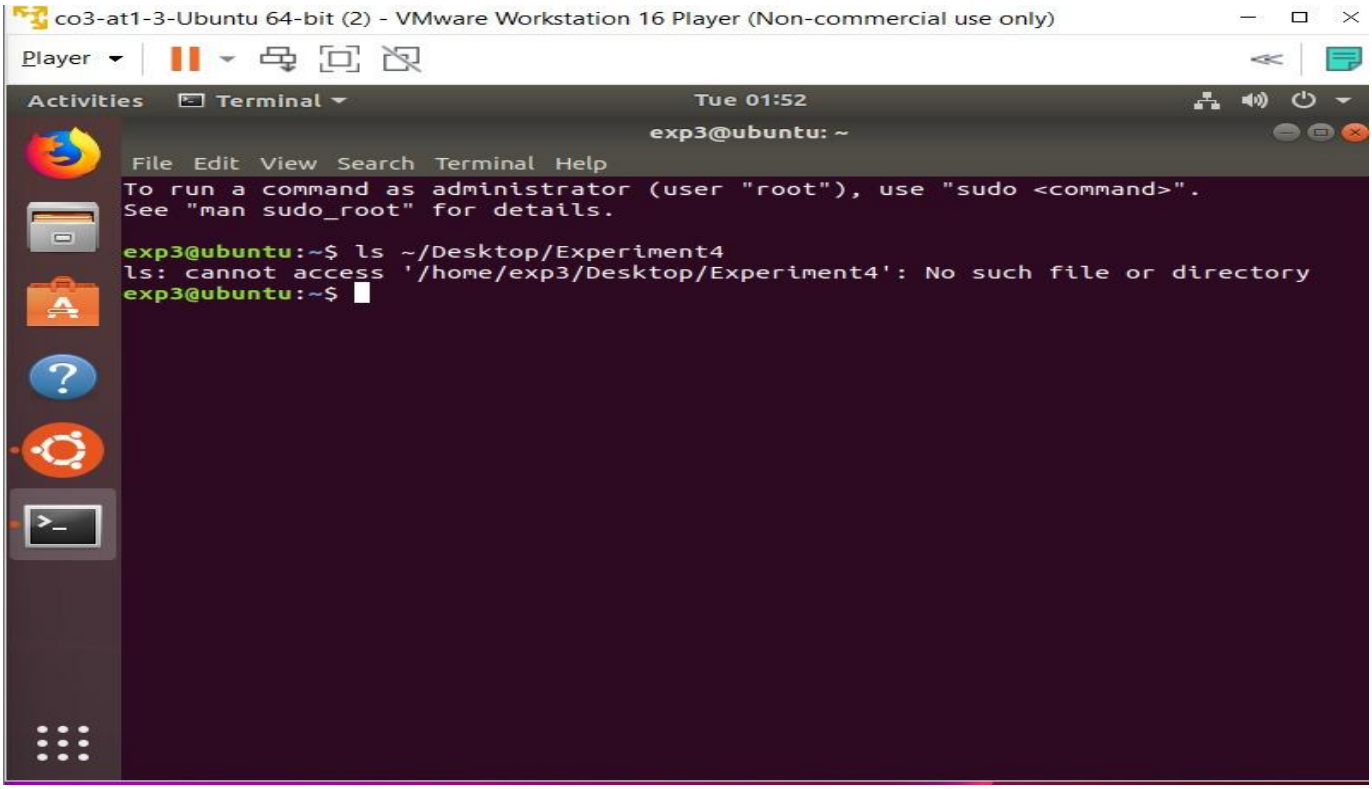
Experiment 4: Create and Test a Virtual Machine Snapshot

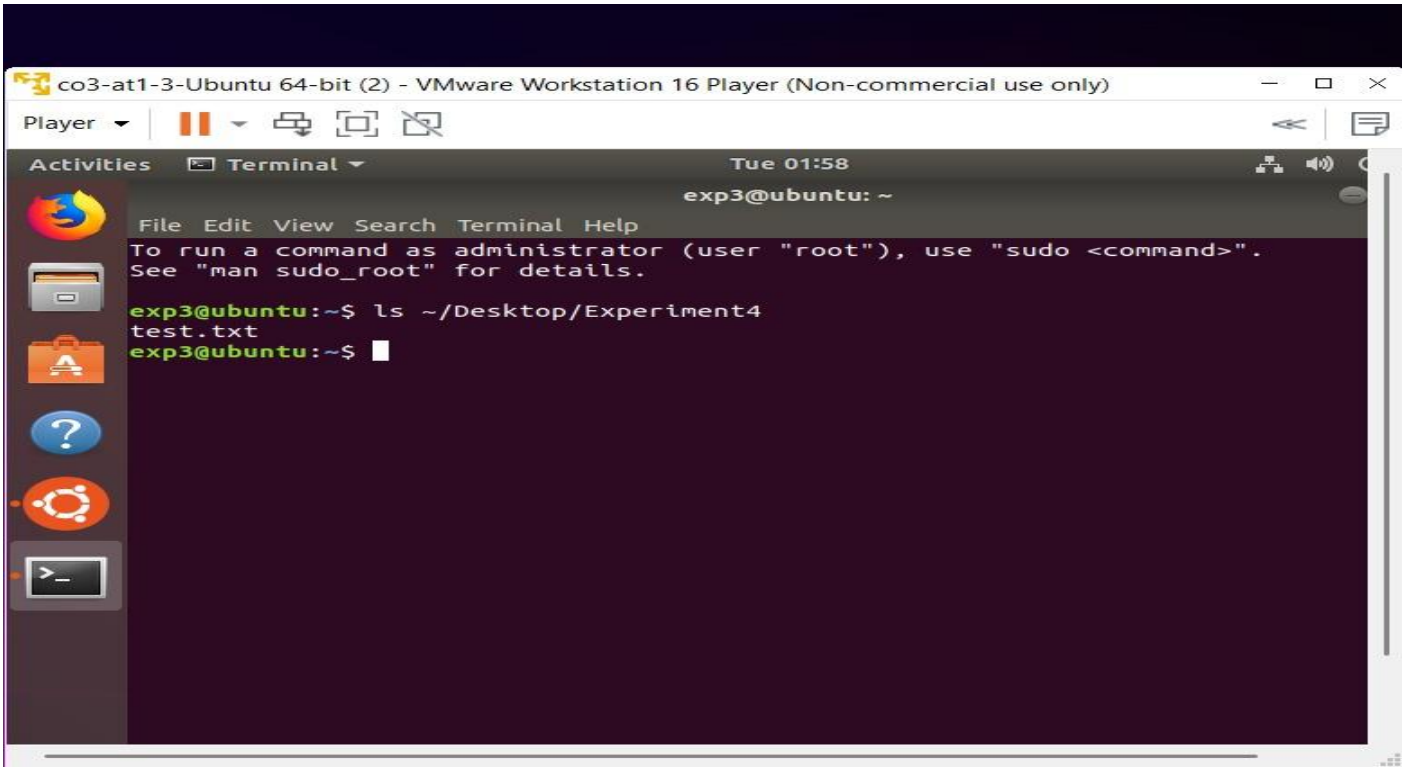
Aim:

To create a snapshot of a virtual machine and restore the previous state using a Type-2 hypervisor.

Algorithm:

1. Open VMware Workstation Pro and start the virtual machine.
2. Create an initial state by adding a test file or making a small change.
3. Shut down the VM if required.
4. Open the VM → Snapshot menu and select Take Snapshot.
5. Enter a snapshot name and save it.
6. Make changes to the virtual machine.
7. Restore the previously created snapshot.
8. Start the VM and verify that the previous state is restored.





Result:

Thus, a virtual machine snapshot was created and restored successfully, and the previous state was verified.